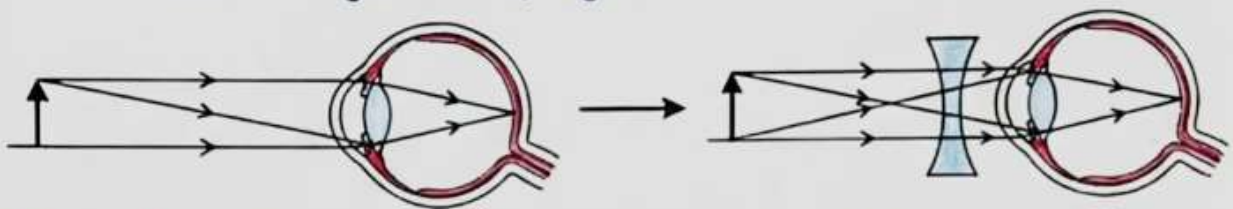


## Defects of Vision

- **Definition:** When the eye cannot see objects clearly at all distances due to some defects in the eye. [Defects of Vision]

### Myopia (Short-sightedness)

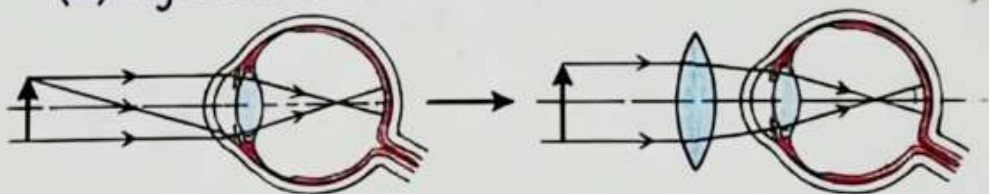
- **Definition:** Can see nearby objects clearly but not distant objects.
- **Causes:** (1) Excessive curvature of eye lens.  
(2) Elongation of eyeball.



- **Correction:** Using a concave lens of suitable power.

### Hypermetropia (Long-sightedness)

- **Definition:** Can see distant objects clearly but not nearby objects.
- **Causes:** (1) Focal length of eye lens is too long.  
(2) Eyeball is too small.



- **Correction:** Using a convex lens of suitable power.

### Presbyopia

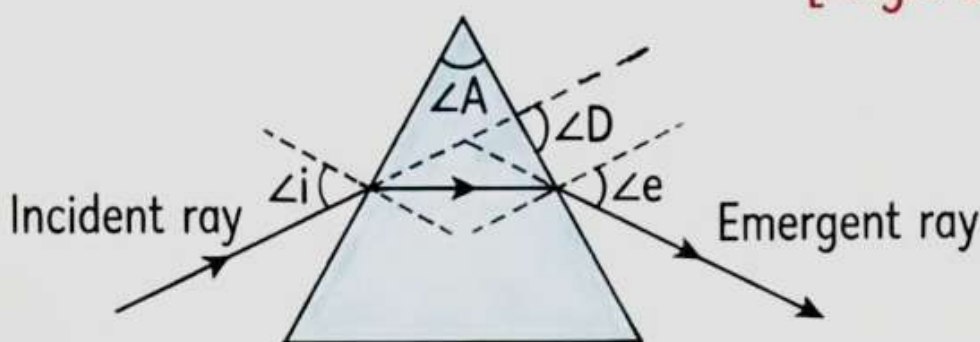
- **Definition:** Decrease in power of accommodation of eye due to ageing.
- **Cause:** Weakening of ciliary muscles and diminishing flexibility of eye lens.
- **Correction:** Using a bifocal lens (both concave and convex).

## Refraction of Light & Dispersion

- **Definition:** The bending of light when it passes through a prism. [Refraction Through Prism]

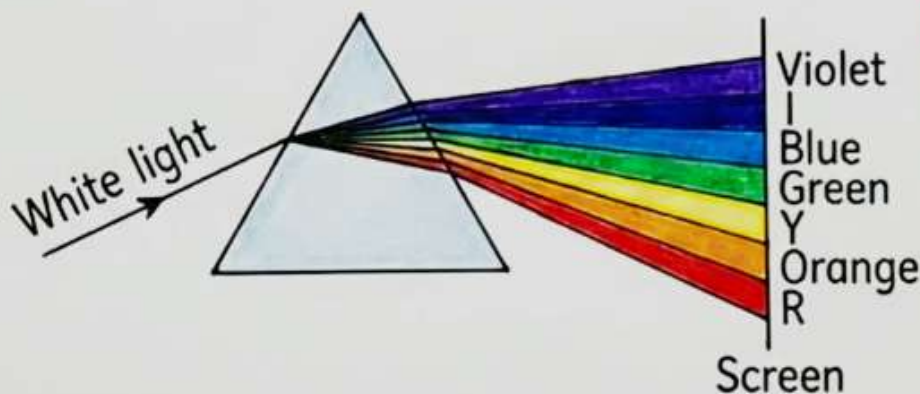
### Refraction Through Prism

- **Deviation of Light:** The angle between the incident ray and the emergent ray is called the angle of deviation ( $\angle D$ ). [Angle of Deviation]



### Dispersion of Light

- **Definition:** The splitting of white light into its constituent colours. [Dispersion]
- **Spectrum:** The band of seven colours obtained on a screen. [Spectrum]



- **VIBGYOR Sequence:** Violet (deviates most), Indigo, Blue, Green, Yellow, Orange, Red (deviates least).
- **Cause of Dispersion:** Different colours of light have different speeds in a medium, so they bend by different amounts. Violet bends the most, Red the least.

# Scattering of Light & Natural Phenomena

## Scattering of Light

- **Definition:** The change in direction of light on striking the particles of the medium. [Scattering of Light]
- **Depends on wavelength:** Scattering is inversely proportional to the fourth power of wavelength ( $1/\lambda^4$ ). Blue scatters more than red. [Wavelength]

## Why is the Sky Blue?

- **Scattering of shorter wavelengths:** The fine particles in the atmosphere scatter blue light (shorter wavelength) the most, making the sky appear blue. [Blue light]

## Why Sun Appears Red at Sunrise and Sunset

- **Longer path of sunlight:** At sunrise/sunset, sunlight travels a longer distance. Most blue light is scattered away.
- **Scattering of blue light:** Only longer wavelengths (red, orange) reach our eyes. [Red light]

## Tyndall Effect

- **Definition:** The scattering of light by colloidal particles in a solution or air, making the path of light visible.
- **Examples:** Sunlight passing through a dusty room, light from a projector in a cinema hall.

- **Definition:** The bending of light on passing through Earth's atmosphere due to changing refractive index. [Atmospheric Refraction]

## Twinkling of Stars

- **Cause:** Continuous change in the atmosphere's refractive index makes the star's apparent position fluctuate. [Changing refractive index]
- **Why planets do not twinkle:** Planets are extended sources, not point sources.

## Advance Sunrise & Delayed Sunset

- **Reason:** Atmospheric refraction bends sunlight, making the sun visible when it's below the horizon. [Refraction]
- **Role:** Makes the sun appear ~2 mins before actual sunrise and after sunset.

## Important Differences

|            |                    |                      |
|------------|--------------------|----------------------|
| Feature    | Myopia             | Hypermetropia        |
| Defect     | Short-sighted      | Long-sighted         |
| Correction | Concave lens       | Convex lens          |
| Feature    | Dispersion         | Scattering           |
| Process    | Splitting of light | Change in direction  |
| Cause      | Different speeds   | Particle interaction |

## HOTS / Exam Points

- **Why danger signals are red:** Red light has the longest wavelength and is scattered the least.
- **Why sky appears dark in space:** No atmosphere to scatter light.